Coupling the field and the point model at the Gauss points

Keisuke Nishioka, meeting on 19 October 2026

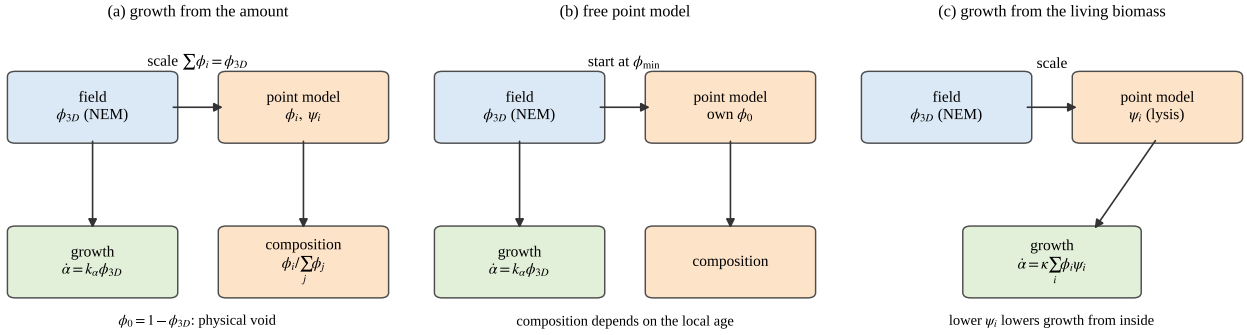
Where I am. The thesis is written; I plan to submit in mid-November (colloquium 16 December). The point model is the Hamilton point model of Klempt et al., with the interaction parameters calibrated by Bayesian inference (TMCMC) on the in-vitro data of Heine et al.; it is not machine-learning based.

1. ϕ_{3D} at the quadrature points. The element solves the field by the NEM directly at the Gauss points and the surface points, so ϕ_{3D} is available at each Gauss point without interpolation. The point model is solved at every Gauss point next to the field.

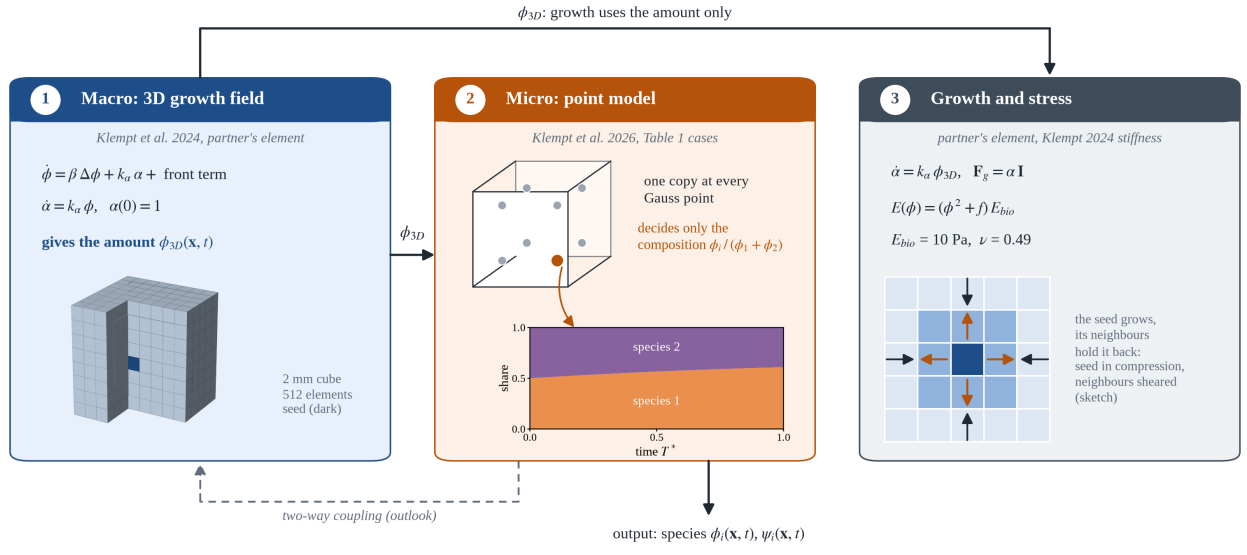
2. The two voids: two schemes implemented. (The thesis calls (a) and (b) the scaled and the unscaled scheme.)

- *(a) Growth from the amount.* The field sets the amount, the point model only the composition $\phi_i / \sum_j \phi_j$ and ψ_i. At each Gauss point and substep the point-model state is scaled to $\sum_i \phi_i = \phi_{3D}$, advanced and scaled again. This sets $\phi_0 = 1 - \phi_{3D}$, i.e. it gives ϕ_0 a physical meaning. When the field's amount equals the point model's own, the composition of the point model alone is recovered (to 10^{-5} or better).
- *(b) Free point model.* ϕ_0 stays auxiliary: the point model runs on its own from the moment the field reaches a threshold and supplies only the composition.

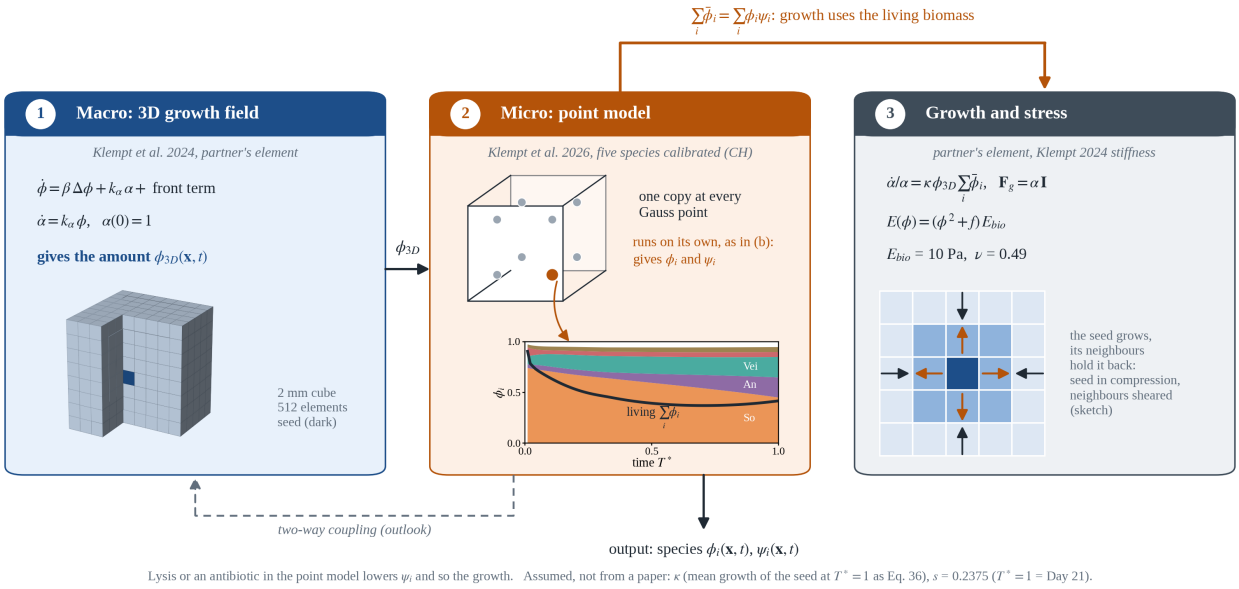
The composition then depends on the local amount (a) or on the local age of the biofilm (b).



Coupling at a Gauss point: (a) growth from the amount, (b) free point model, (c) growth from the living biomass of the point model.



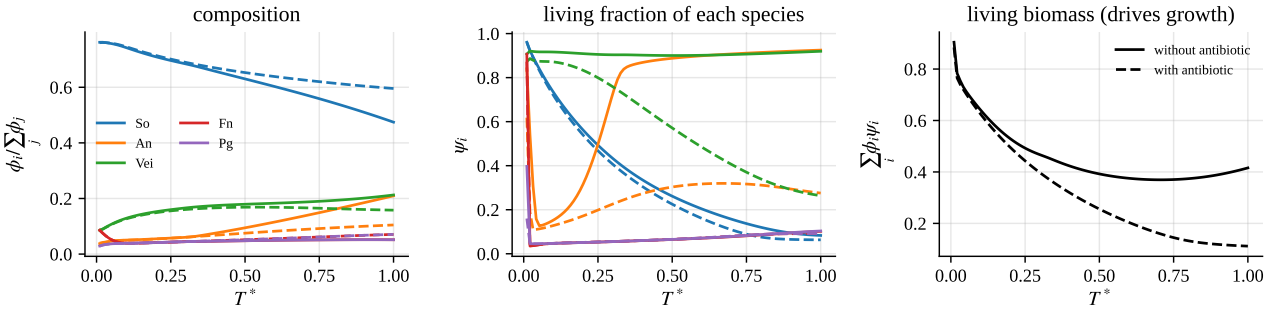
Scheme (a), as in the thesis (two species shown): the field gives the amount ϕ_{3D}, the point model at every Gauss point only the composition, and growth and stress use ϕ_{3D}.



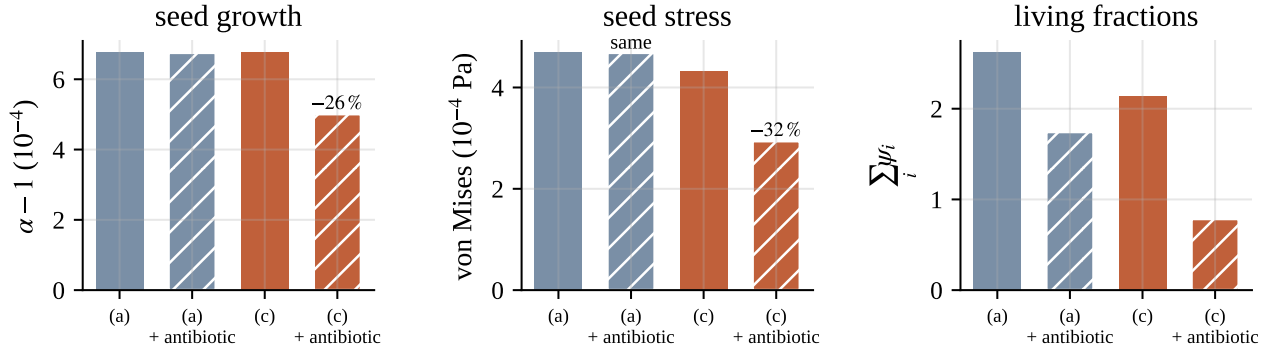
Lysis or an antibiotic in the point model lowers ψ_i and so the growth. Assumed, not from a paper: κ (mean growth of the seed at $T^* = 1$ as Eq. 36), $s = 0.2375$ ($T^* = 1$ = Day 21).

Scheme (c): the point model also gives the living fractions ψ_i, and growth uses the living biomass $\sum_i \bar{\phi}_i = \sum_i \phi_i \psi_i$ (inset: calibrated commensal flow run, seed Gauss point).

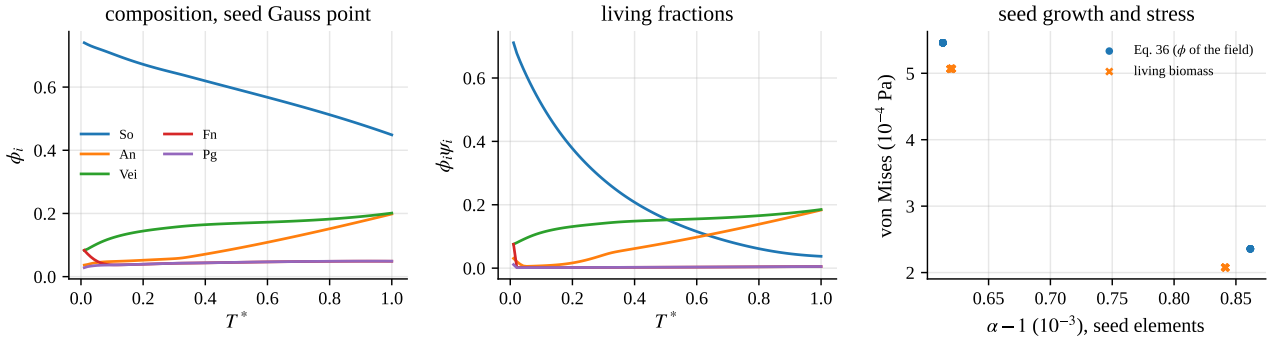
3. Lysis acting on growth. In a first run with the calibrated five-species model, α grows with the living biomass $\sum_i \phi_i \psi_i$ of the point model times the local amount: $\dot{\alpha}/\alpha = \kappa \phi_{3D} \sum_i \bar{\phi}_i$, with $\bar{\phi}_i = \phi_i \psi_i$ and the point model running on its own as in (b). A drop of ψ_i (lysis, or an antibiotic in the point model) then lowers the growth from inside the point model. With the rate chosen so that the mean growth equals that of Eq. 36, the mean von Mises stress of the seed is 7 % lower (−7.0 % at 8^3, −7.1 % at 16^3, with the calibration to the measured data; −7.8 % with the earlier calibration). With an antibiotic in the point model (same sensitivity for all five species, strength chosen so that the living fraction halves by $T^* = 1$; both assumed), scheme (c) gives 26 % less growth and 32 % less seed stress, while scheme (a) gives the same stress as without the antibiotic, only the composition changes. The growth and the stress are proportional to κ. The rates are my choice, not from a paper.



Point model at a seed Gauss point, scheme (c), without (solid) and with (dashed) the antibiotic, applied at a constant level from $T^* = 0$.

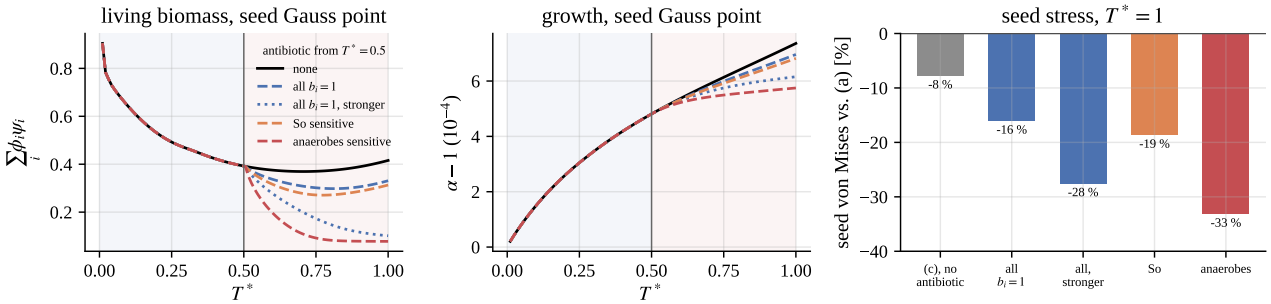


Antibiotic in the point model (commensal flow condition, 8^3, $T^* = 1$, means over the seed elements): (a) growth from ϕ of the field, (c) growth from the living biomass.



Calibrated five-species model in the element (commensal flow condition, 8^3, $T^* = 1$): composition and living fractions at a seed Gauss point; growth and stress of the seed elements for Eq. 36 and for growth from the living biomass.

Antibiotic from $T^* = 0.5$. As in the validation case of Fritsch et al. (2025, Fig. 18), the antibiotic is switched on at $T^* = 0.5$. Up to $T^* = 0.5$ the run equals the one without antibiotic; after it the growth slows. With the same sensitivity for all species the seed stress is 16 % lower than in scheme (a) (28 % with a stronger antibiotic); with the anaerobes (*V. dispar*, *F. nucleatum*, *P. gingivalis*) ten times more sensitive it is 33 % lower, with only *S. oralis* sensitive 19 %. Scheme (a) gives the base stress in all four cases.



Antibiotic switched on at $T^* = 0.5$, scheme (c), commensal flow condition, 8^3. Left and middle: first seed Gauss point. Right: mean von Mises stress of the seed elements relative to scheme (a). Sensitivities and strengths are assumed.

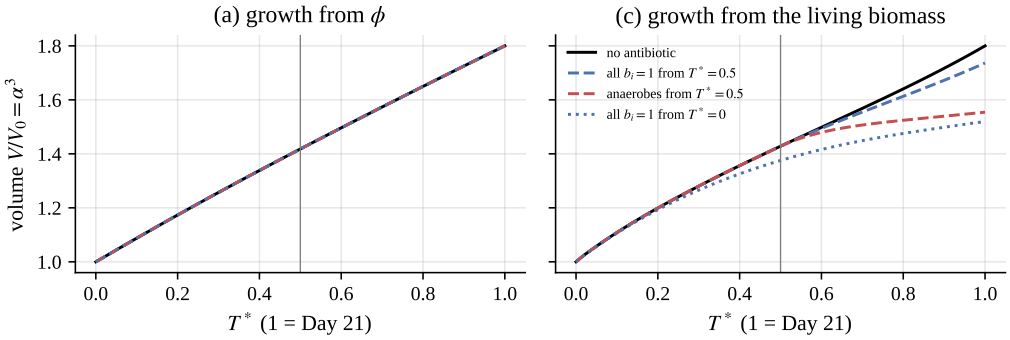
Comparison and proposal.

	(a) growth from the amount	(b) free point model	(c) growth from the living biomass
What it does	amount from the field, composition from the point model, scaled to $\sum_i \phi_i = \phi_{3D}$	point model runs on its own, supplies the composition only	point model as (b); growth $\dot{\alpha}/\alpha = \kappa\,\phi_{3D}\sum_i \phi_i \psi_i$
Meaning of ϕ_0	physical void, $1 - \phi_{3D}$	auxiliary, as in the paper	as (b)
Composition depends on	local amount	local age of the biofilm	local age of the biofilm
Lysis acts on growth	no	no	yes, lower ψ_i lowers growth
Check	recovers the point model alone to 10^{-5}	point-model trajectory, shifted in time	8^3, 16^3: seed stress -7 %; with antibiotic -32 %, (a) unchanged
In the thesis	Chapter 4	Chapter 4, for comparison	Outlook, one paragraph
Strength	well tested, simple to explain	closest to the paper's reading of ϕ_0	answers the lysis question; growth from the producing biomass has precedent in the literature
Weakness	gives ϕ_0 a physical meaning (stated as an assumption)	nearly decoupled from the field; cannot feed lysis back	κ is my choice; 8^3 only so far

Proposal: (a) as the basis of the thesis, (b) as the comparison for the interpretation of ϕ_0, and (c), which keeps the point model of (b) and lets its living biomass drive the growth, as the extension that lets lysis act on growth.

Which of (a) and (c) is right? This is open. Eq. 36 of Klempt et al. 2024 is written for one species without dead cells, so it does not say whether the growth should follow all of ϕ or only the living part. Biologically (c) is more natural: the volume grows through division and matrix production of living cells, and classical biofilm

models (Wanner and Gujer 1986; Alpkvist and Klapper 2007) let only the active biomass grow. The volume data of Heine et al. cannot decide it: in the commensal flow condition the volume keeps growing while the live fraction drops to 0.35–0.7, which both schemes can follow. Fitted to the growing part of the measured volumes, with the measured live fraction for (c) and the same number of parameters, (c) fits slightly better in all four comparisons, but the difference is small ($\Delta\chi^2 \leq 0.24$). Feng et al. (2021) also let each species grow with its own amount. Experiment 1 below would decide it: with the rate fitted to the measured volume ($\times 1.80$ from Day 3 to Day 21), an antibiotic on the anaerobes from half time leaves (a) at 1.80 and lowers (c) to 1.55.



Predicted volume of the commensal flow biofilm, from the point model at a seed Gauss point, growth rate fitted to the measured volume; mechanics not computed. Antibiotic parameters assumed.

4. Limits stated in the thesis. The coupling is one-way for the mechanics (the composition does not change the stiffness). Scheme (a) gives ϕ_0 a meaning the point model does not intend. A coupling without these choices would need a spatial multispecies model from one variational principle. A recent check: the value of ϕ at the surface points is a mean of neighbouring Gauss points, exact only for constant fields, so the Laplacian next to a face has an error that does not decrease with the mesh; in my seed runs this layer carries 0.1 to 0.2 % of the error.

Experiments that could test the models (ideas only).

1. *Two antibiotics during growth.* Add an antibiotic at a constant concentration to the five-species biofilm of Heine et al. partway through (as in the validation case of Fritsch et al. 2025, Fig. 18[1]), once a broad one (chlorhexidine) and once one that acts on the anaerobes (metronidazole: *V. dispar*, *F. nucleatum*, *P. gingivalis*, less on *S. oralis* and *A. naeslundii*); measure the live fraction (LIVE/DEAD) and the biofilm volume (CLSM). Scheme (c) predicts that the volume growth slows when the dominant species is hit, so differently for the two drugs and for the commensal and the dysbiotic community; scheme (a) predicts no change of the volume for either. This decides between them.
2. *Co-culture without Fn.* Grow the dysbiotic community under flow without *F. nucleatum*. The calibrated model predicts that the late increase of *P. gingivalis* disappears.
3. *Spatial composition.* Depth profiles of the species by FISH and CLSM (partly available at NIFE). Composition set by the local amount (a) or by the local age (b) gives different profiles; this bears on the meaning of ϕ_0.
4. *Stiffness.* Young's modulus by AFM per condition or per community (as in Pattem et al. 2018), replacing the assumed $E = 10$ Pa; a dependence on the composition would support a two-way coupling.
5. *Growth-induced deformation.* Biofilm grown on a soft substrate (gel or thin membrane); the measured deformation gives the growth-induced stress, the closest available check of the stresses.

These are ideas for the experimental partners; I return to Japan after the colloquium and would not run them myself. Item 1 would decide between (a) and (c).

To decide together.

Meaning of ϕ_0. I would keep scheme (a) as the basis and state $\phi_0 = 1 - \phi_{3D}$, the void of the field, as an assumption, with scheme (b) as a comparison. It is the simpler reading to explain. Scheme (c) keeps ϕ_0

[1] L. Fritsch, H. Geisler, J. Grashorn, F. Klempt, M. Soleimani, M. Broggi, P. Junker, M. Beer: Bayesian updating of constitutive parameters under hybrid uncertainties with a novel surrogate model applied to biofilms. arXiv:2512.15145, 2025.

auxiliary as in the paper (its point model runs as in (b)), so the continuation would use the paper's reading. Do you agree with this split, or would you rather keep ϕ_0 auxiliary in the thesis too?